

MAYSB01817

## Methyl propiolate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 丙炔酸甲酯  
 Methyl propiolate

**Cat No. :** SB01817DA; SB01817EA; SB01817EB; SB01817FL; SB01817ZZ  
**Synonyms** Propynoic acid, methyl ester  
**CAS No** 922-67-8  
**Molecular Formula** C4 H4 O2

**Supplier**
**UK entity/business name**  
 Thermo Fisher Scientific (Heysham),  
 Shore Road,  
 Port of Heysham Industrial Park,  
 Heysham, Lancashire, LA3 2XY  
 United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a  
 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
Odorless

#### Emergency Overview

Highly flammable liquid and vapor. Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation.  
 Lachrymator (substance which increases the flow of tears).

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Flammable liquids.                                 | Category 2 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P243 - Take action to prevent static discharges  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P312 - Call a POISON CENTER or doctor if you feel unwell  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Highly flammable. Vapors may cause flash fire or explosion.

**Health Hazards**

Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation. Lachrymator (substance which increases the flow of tears).

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility. The product is insoluble and floats on water. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil.

**Other Hazards**

Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No   | Weight % |
|-------------------|----------|----------|
| Methyl propiolate | 922-67-8 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

None reasonably foreseeable. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Flammable. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## Methyl propiolate

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage**

Keep away from heat, sparks and flame. Flammables area. Keep refrigerated. Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**SAFETY DATA SHEET****Methyl propiolate**

|                                   |                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Large scale/emergency use</b>  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced |
| <b>Small scale/Laboratory use</b> | Maintain adequate ventilation                                                                                                                       |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                                |                                |                                             |
|------------------------------------------------|--------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless                      |                                             |
| <b>Physical State</b>                          | Liquid                         |                                             |
| <b>Odor</b>                                    | Odorless                       |                                             |
| <b>Odor Threshold</b>                          | No data available              |                                             |
| <b>pH</b>                                      | No information available       |                                             |
| <b>Melting Point/Range</b>                     | No data available              |                                             |
| <b>Softening Point</b>                         | No data available              |                                             |
| <b>Boiling Point/Range</b>                     | 98 - 101 °C / 208.4 - 213.8 °F | @ 760 mmHg                                  |
| <b>Flash Point</b>                             | 10 °C / 50 °F                  | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available              |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                 | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available              |                                             |
| <b>Vapor Pressure</b>                          | No data available              |                                             |
| <b>Vapor Density</b>                           | 2.90                           | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.940                          |                                             |
| <b>Bulk Density</b>                            | Not applicable                 | Liquid                                      |
| <b>Water Solubility</b>                        | Insoluble                      |                                             |
| <b>Solubility in other solvents</b>            | No information available       |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                |                                             |
| <b>Autoignition Temperature</b>                | No data available              |                                             |
| <b>Decomposition Temperature</b>               | No data available              |                                             |
| <b>Viscosity</b>                               | No data available              |                                             |
| <b>Explosive Properties</b>                    |                                | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available       |                                             |
| <b>Molecular Formula</b>                       | C4 H4 O2                       |                                             |
| <b>Molecular Weight</b>                        | 84.07                          |                                             |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under recommended storage conditions.                                                                      |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                          |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. |
| <b>Materials to avoid</b>               | Acids. Bases. Strong oxidizing agents.                                                                            |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                                          |

**SECTION 11. TOXICOLOGICAL INFORMATION**

|                                                   |                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | The toxicological properties have not been fully investigated                                                                 |
| <b>(a) acute toxicity;</b>                        |                                                                                                                               |
| <b>(b) skin corrosion/irritation;</b>             | Category 2                                                                                                                    |
| <b>(c) serious eye damage/irritation;</b>         | Category 2                                                                                                                    |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                               |
| <b>Respiratory</b>                                | Based on available data, the classification criteria are not met                                                              |
| <b>Skin</b>                                       | Based on available data, the classification criteria are not met                                                              |
| <b>(e) germ cell mutagenicity;</b>                | Based on available data, the classification criteria are not met                                                              |
| <b>(f) carcinogenicity;</b>                       | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product |
| <b>(g) reproductive toxicity;</b>                 | Based on available data, the classification criteria are not met                                                              |
| <b>(h) STOT-single exposure;</b>                  | Category 3                                                                                                                    |
| <b>Results / Target organs</b>                    | Respiratory system                                                                                                            |
| <b>(i) STOT-repeated exposure;</b>                | Based on available data, the classification criteria are not met                                                              |
| <b>Target Organs</b>                              | None known.                                                                                                                   |
| <b>(j) aspiration hazard;</b>                     | Based on available data, the classification criteria are not met                                                              |
| <b>Other Adverse Effects</b>                      | No information available                                                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting           |

**SECTION 12. ECOLOGICAL INFORMATION**

|                                        |                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>             | Do not empty into drains.                                                                                                                                                                                                                                                                                                |
| <b>Persistence and Degradability</b>   |                                                                                                                                                                                                                                                                                                                          |
| <b>Persistence</b>                     | Insoluble in water, Persistence is unlikely, based on information available.                                                                                                                                                                                                                                             |
| <b>Bioaccumulative Potential</b>       | May have some potential to bioaccumulate                                                                                                                                                                                                                                                                                 |
| <b>Mobility in soil</b>                | The product is insoluble and floats on water The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility Will likely be mobile in the environment due to its volatility |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                                                                                                                                                                                |

**SAFETY DATA SHEET****Methyl propiolate****Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN3272  
Proper Shipping Name Esters, n.o.s  
Technical Shipping Name Methyl propiolate  
Hazard Class 3  
Packing Group II

**IMDG/IMO**

UN-No UN3272  
Proper Shipping Name Esters, n.o.s  
Technical Shipping Name Methyl propiolate  
Hazard Class 3  
Packing Group II

**IATA**

UN-No UN3272  
Proper Shipping Name Esters, n.o.s  
Technical Shipping Name Methyl propiolate  
Hazard Class 3  
Packing Group II

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Methyl propiolate | -                                                   | -                                       | X    | X     | 213-083-5 | -    | -   | -     | -    | X    | -    | KE-25290 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 08-Aug-2007  
**Revision Date** 01-Sep-2023  
**Revision Summary** 9, 11, 12, 15, 16, SDS sections updated, 1, 2.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**